

Notes of HDP¹

Yue Wu

August 4, 2026

¹Based on IEOR8100 High-dimensional Probability with Applications at Columbia University, taught by Prof. Kaizheng Wang

Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian, weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian, strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema

- Finite Maxima
- The Covering Method
- The Chaining Method

Learning Theory

Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian, weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian, strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema

- Finite Maxima
- The Covering Method
- The Chaining Method

Learning Theory

Intro

Informal version of

- ▶ Concentration
- ▶ Suprema

Preliminaries and Notations

- ▶ $a \lesssim b$: $a \leq Cb$ where C is some absolute constant.
- ▶ -[Jensen inequality] for any random variable X and a convex function $f : \mathbb{R} \rightarrow \mathbb{R}$, we have

$$f(\mathbb{E} X) \leq \mathbb{E} f(X).$$

More generally, this holds for any random vector X taking values in $\mathbb{R}^n$ and any convex function $f : \mathbb{R}^n \rightarrow \mathbb{R}$.

Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian, weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian, strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema

- Finite Maxima
- The Covering Method
- The Chaining Method

Learning Theory

Intro - Concentration inequalities

(1) Variance version of concentration

$$\text{Var}(f) \lesssim \sum_{i=1}^n \mathbb{E} [(D_i f)^2]$$

the variance of a function f of independent (or weakly dependent) random variables is small if the “gradient” of f is small.

(2) Sub-Gaussianity, a sharper control on the distribution of the fluctuations

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \left\| \sum_{i=1}^n D_i f \right\|_{\infty}^2 \quad \text{by simple martingale}$$

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \left\| \sum_{i=1}^n D_i f \right\|_{\infty}^2 \quad \text{by entropy method}$$



(3)

We will repeatedly encounter this idea in the setting of concentration inequalities: typically the expectation of the “square gradient” controls the variance, while a uniform bound on the “square gradient” controls the subgaussian variance proxy.’

Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian, weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian, strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema

- Finite Maxima
- The Covering Method
- The Chaining Method

Learning Theory

Tensorization property of the Variance

Efron-Stein

- relationship between the variance of the overall function and the sum of the individual expected variance[] contributions

[Efron-Stein] Let $X_1, \dots, X_n$ be independent random variables, define

$$\text{Var}_i f(x_1, \dots, x_n) := \text{Var}[f(x_1, \dots, x_{i-1}, \mathbf{x}_i, x_{i+1}, \dots, x_n)] = \text{Var}(f \mid X_{-i})$$

Then

$$\text{Var}[f(X_1, \dots, X_n)] \leq \sum_{i=1}^n \mathbb{E}[\text{Var}_i f(X_1, \dots, X_n)]$$

Bounded Difference Inequalities

Variance version

- relationship between variance of overall function and the sum of Difference $D_i f$ with respect to individual variable

[Bounded Difference Inequalities] Let $X_1, \dots, X_n$ be independent random variables, define

$$D_i f(x_1, \dots, x_n) = \sup_{x_i} f - \inf_{x_i} f$$

$$D_i^- f(x_1, \dots, x_n) = f - \inf_{x_i} f$$

Then

$$\text{Var}(f) \leq \frac{1}{4} \sum_{i=1}^n \mathbb{E} [(D_i f)^2]$$

$$\text{Var}(f) \leq \frac{1}{4} \sum_{i=1}^n \mathbb{E} [(D_i^- f)^2]$$

Example:

$D_i f$, a 'discrete derivative' of the function with respect to the i -th variable

Poincaré Inequalities

- ▶ from 'variance/difference' to 'derivative'

$$\text{Var}(f) \lesssim \mathbb{E} [\|\text{gradient}(f)\|^2]$$

[Convex Poincaré Inequalities] Let $\{X_i\}_{i=1}^n$ be independent random variables, $X_i \in [a, b]$, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ convex and differentiable. Then

$$\text{Var}(f) \leq (b - a)^2 \cdot \mathbb{E} \|\nabla f(X_1, \dots, X_n)\|_2^2$$

proof: first relate 'difference' with 'derivative' with respect to certain assumption on X and f , then use efron-stein/bounded difference inequalities

Poincaré Inequalities

General version

Poincaré Inequalities

Application to Sampling

Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian, weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian, strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema

- Finite Maxima
- The Covering Method
- The Chaining Method

Learning Theory

Intro

Subgaussian Concentration

- ▶ Subgaussianity: another way to describe concentration
- ▶ Entropy Method
- ▶ Log-Sobolev Inequalities

Tail probability bounded by 'Moments'

Chebyshev inequality, Chernoff bound

Let X be a random variable. Then, for any $t > 0$, we have

- ▶ -[Chebyshev inequality]

$$\mathbb{P}\{|X - \mathbb{E}X| \geq t\} \leq \frac{\mathbb{E}|X - \mathbb{E}X|^p}{t^p}$$

- ▶ -[Chernoff bound, exponential moment method] Define the log-moment generating function ψ of a random variable X and its Legendre dual ψ^* as

$$\psi(\lambda) := \log \mathbb{E}e^{\lambda(X - \mathbb{E}X)}, \quad \psi^*(t) = \sup_{\lambda \geq 0} \{\lambda t - \psi(\lambda)\}.$$

Then

$$\mathbb{P}\{X - \mathbb{E}X \geq t\} \leq e^{-\psi^*(t)} \quad \text{for all } t \geq 0.$$

bounding tail probability $\rightarrow$ bounding the MGF.

proof: by Markov Inequality on nonnegative increasing function of $X - \mathbb{E}X$

Subgaussian Variable

[Subgaussian properties] Let X be a random variable. The follows are equivalent, with $K_i > 0$ differing by at most an absolute constant.

(i) (Tail bound) There exists $K_1 > 0$ such that

$$\mathbb{P}\{|X| \geq t\} \leq 2 \exp(-t^2/K_1^2) \quad \text{for all } t \geq 0.$$

(ii) (Moment growth bound) There exists $K_2 > 0$ such that

$$\frac{\|X\|_{L^p}}{\sqrt{p}} = \frac{(\mathbb{E}|X|^p)^{1/p}}{\sqrt{p}} \leq K_2 \quad \text{for all } p \geq 1.$$

[Subgaussian norm] $\|X\|_{\psi_2} = \sup_{p \geq 1} \left\{ \frac{(\mathbb{E}|X|^p)^{1/p}}{\sqrt{p}} \right\}$, the smallest K_2 .

(iii) (Finite MGF of X^2) There exists $K_3 > 0$ such that

$$\mathbb{E} \exp(X^2/K_3^2) \leq 2.$$

Moreover, if $\mathbb{E}X = 0$, (i)–(iii) are equivalent to the following one:

(iv) (MGF bound) There exists $K_4 > 0$ such that

$$\mathbb{E} \exp(\lambda X) \leq \exp(\lambda^2 K_4^2) \quad \text{for all } \lambda \in \mathbb{R}.$$

[Centering] Any subgaussian random variable X satisfies

$$\|X - \mathbb{E}X\|_{\psi_2} \leq C\|X\|_{\psi_2}.$$

Subgaussianity

a centered version

[Subgaussianity] A random variable X is σ^2 -subgaussian if

$$\mathbb{E}e^{\lambda(X-\mathbb{E}X)} \leq e^{\lambda^2\sigma^2/2}, \quad \forall \lambda \in \mathbb{R}$$

If X is σ^2 -subgaussian, then

$$\mathbb{P}\{X - \mathbb{E}X \geq t\} \leq e^{-t^2/2\sigma^2}, \quad \mathbb{P}\{X - \mathbb{E}X \leq -t\} \leq e^{-t^2/2\sigma^2}.$$

This is also equivalent to

$$\|X - \mathbb{E}X\|_{\psi_2}^2 \lesssim \sigma^2.$$

σ^2 is called the variance proxy because it shows like variance, the sum of n independent σ^2 -subg variable is $\frac{1}{n}\sigma^2$ -subg.

Subgaussianity

What r.v.'s are subgaussian?

- ▶ Gaussian
- ▶ A bounded r.v. [Hoeffding] If $a \leq X \leq b$ a.s. for constants a & b , then X is $\frac{(b-a)^2}{4}$ -subg.

Subgaussianity type of concentration of a function

Bounded Difference Inequalities

Subgaussianity is another description of concentration, and now we discuss the subgaussianity of function $f(X_1, \dots, X_n)$.

[Azuma-Hoeffding inequality] (concentration of the sums of martingale differences)

[McDiarmid] If $\{X_i\}_{i=1}^n$ are indep, $\|D_i f\|_\infty < \infty$ for all i , then $f(X_1, \dots, X_n)$ is subgaussian with variance proxy $\frac{1}{4} \sum_{i=1}^n \|D_i f\|_\infty^2$

$$\mathbb{P}\{f - \mathbb{E}f \geq t\} \leq e^{-2t^2 / \sum_{i=1}^n \|D_i f\|_\infty^2}$$

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \sum_{i=1}^n \|D_i f\|_\infty^2.$$

Actually subgaussian property does not tensorize naturally, and the martingale method can only partially address this issue. Next, we explore the entropy method to derive a better bound.

Entropy and Subgaussian

[Entropy] The entropy of a nonnegative random variable Z is

$$\text{Ent}[Z] := \mathbb{E}[Z \log Z] - \mathbb{E}[Z] \log \mathbb{E}[Z].$$

A sufficient condition for subgaussian in terms of entropy:

[Herbst] If $\text{Ent}(e^{\lambda X}) \leq \frac{\lambda^2 \sigma^2}{2} \mathbb{E} e^{\lambda X}$, $\forall \lambda$, then X is σ^2 -subg.

Tensorization property of the Entropy

Let $X_1, \dots, X_n$ be independent random variables, $f \geq 0$, define

$$g_i(x_1, \dots, x_n) = \text{Ent} [f(x_1, \dots, x_{i-1}, X_i, x_{i+1}, \dots, x_n)]$$

then

$$\text{Ent} [f(X_1, \dots, X_n)] \leq \sum_{i=1}^n \mathbb{E} [g_i(X_1, \dots, X_n)]$$

Bounded Difference Inequalities

a stronger version derived by entropy method

If $\{X_i\}_{i=1}^n$ indep. $\|D_i f\|_\infty < \infty$ for all i , then

$$\|f - \mathbb{E}f\|_{\psi_2}^2 \lesssim \left\| \sum_{i=1}^n |D_i f|^2 \right\|_\infty$$

Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian, weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian, strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema

- Finite Maxima
- The Covering Method
- The Chaining Method

Learning Theory

Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian, weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian, strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema

- Finite Maxima
- The Covering Method
- The Chaining Method

Learning Theory

Intro - Suprema

- ▶ Previous chapters: the concentration of a function $f \rightarrow \mathbb{E}f$
- ▶ This chapter: the magnitude of the mean $\mathbb{E}f$

Suprema: focus on a class of function

$$f = \sup_{t \in T} X_t$$

and discuss the relation of $\mathbb{E}(\sup_{t \in T} X_t)$ with complexity of T .

Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian, weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian, strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema

Finite Maxima

- The Covering Method
- The Chaining Method

Learning Theory

Finite Maxima

Suppose $|T| < \infty$

- ▶ if X_t have bounded first moment,

$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq \sum_{t \in T} \mathbb{E} |X_t| \leq |T| \cdot \sup_{t \in T} \mathbb{E} |X_t|$$

- ▶ if X_t have p -th bounded moment,

$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq \left[\mathbb{E} \left(\sup_{t \in T} |X_t|^p \right) \right]^{\frac{1}{p}} \leq |T|^{\frac{1}{p}} \cdot \sup_{t \in T} (\mathbb{E} |X_t|^p)^{\frac{1}{p}}$$

- ▶ -[Maximal inequality] Suppose that $\log \mathbb{E}[e^{\lambda X_t}] \leq \psi(\lambda)$ for all $\lambda \geq 0$ and $t \in T$, where ψ is convex and $\psi(0) = \psi'(0) = 0$. Then

$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq (\psi^*)^{-1} \log |T|$$

- ▶ if X_t is σ^2 -subg, then

$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq \sqrt{2\sigma^2 \log |T|}$$

- ▶ -[Maximal tail inequality]

Application of finite maxima

Model Selection Wang 2026

Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian, weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian, strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema

Finite Maxima

The Covering Method

The Chaining Method

Learning Theory

The Covering Method

Let T_ε be an ε -cover of T , $\forall t \in T$, $\exists \pi_0(t) \in T_\varepsilon$, s.t. $d(t, \pi_0(t)) \leq \varepsilon$

$$\begin{aligned}\mathbb{E} \left(\sup_{t \in T} X_t \right) &\leq \inf_{\varepsilon > 0} \left\{ \mathbb{E} \left(\sup_{t \in T} X_{\pi_0(t)} \right) + \mathbb{E} \left(\sup_{t \in T} |X_t - X_{\pi_0(t)}| \right) \right\} \\ &\leq \inf_{\varepsilon > 0} \left\{ \mathbb{E} \left(\sup_{t \in T} X_{\pi_0(t)} \right) + \varepsilon \mathbb{E} C \right\} \quad (\text{C-Lip process}) \\ &\leq \inf_{\varepsilon > 0} \left\{ (\psi^*)^{-1} \log |T_\varepsilon| + \varepsilon \mathbb{E} C \right\} \quad (\text{finite maxima})\end{aligned}$$

How to bound $\mathbb{E} \left(\sup_{t \in T} X_t \right)$ if $|T|$ is infinite? **Discretization.**

- ▶ We need the continuity of the random process $\{X_t\}_{t \in T}$. [Lipschitz]
- ▶ And find finite representatives of X_t . How large is the finite representatives? [covering number]
- ▶ so that for arbitrary $t \in T$ we can represent $\sup$
- ▶ apply the finite maximal ineq. to an optimal ε -net.

Covering Number

[Covering number] Consider a metric space (T, d)

- A finite set $T_\varepsilon \subset T$ is called an ε -**net** (also called ε -cover) of T if (every point in T_ε is within distance ε of some point of T)

$$\forall t \in T, \exists \pi(t) \in T_\varepsilon \text{ s.t. } d(t, \pi(t)) \leq \varepsilon.$$

- The smallest cardinality of an ε -net for (T, d) is called the **covering number**. (a measure of the complexity of the set T at the scale ε)

$$\mathcal{N}(T, d, \varepsilon) := \inf\{|T_\varepsilon|\}.$$

[Metric entropy] $\log \mathcal{N}(T, d, \varepsilon)$

[Covering number of unit ball] $B_d(x, r) := \{y \in \mathbb{R}^d : \|y - x\|_2 \leq r\}.$

Then,

$$\left(\frac{1}{\varepsilon}\right)^d \leq \mathcal{N}(B_d(0, 1), \|\cdot\|_2, \varepsilon) \leq \left(1 + \frac{2}{\varepsilon}\right)^d, \quad \forall \varepsilon > 0.$$

Continuity of a random process

Lipschitz process

[Lipschitz process] The random process $\{X_t\}_{t \in T}$ is called C -Lipschitz for a metric d on T if there exists a random variable C such that

$$|X_t - X_s| \leq Cd(t, s) \quad \text{for all } t, s \in T.$$

Example:

- ▶ Lipschitz models
- ▶ Linear class

Lipschitz Maximal Inequality

[Lipschitz Maximal Inequality]

Suppose that $\{X_t\}_{t \in T}$ is C -Lip and $\log \mathbb{E}[e^{\lambda X_t}] \leq \psi(\lambda)$ for all $\lambda \geq 0$ and $t \in T$, where ψ is convex and $\psi(0) = \psi'(0) = 0$. Then,

$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq \inf_{\varepsilon > 0} \left\{ (\psi^*)^{-1} [\log \mathcal{N}(T, d, \varepsilon)] + \varepsilon \cdot \mathbb{E} C \right\}.$$

► if X_t is σ^2 -subg, then

$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq \inf_{\varepsilon > 0} \left\{ \sqrt{2\sigma^2 [\log \mathcal{N}(T, d, \varepsilon)]} + \varepsilon \cdot \mathbb{E} C \right\}.$$

Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian, weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian, strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema

- Finite Maxima
- The Covering Method
- The Chaining Method**

Learning Theory

The Chaining Method

How to bound $\mathbb{E} \left(\sup_{t \in \mathcal{T}} X_t \right)$ if $|\mathcal{T}|$ is infinite? And the random process $\{X_t\}_{t \in \mathcal{T}}$ is Lipschitz "in probability"?

The one-resolution discretization may give us sub-optimal bounds of the generalization gap.

Continuity of a random process

ψ process, separable process

Dudley

multi-resolution approximation

[Dudley] Let $\{X_t\}_{t \in T}$ be a separable Ψ -proc. on a metric space (T, d) .
Then,

$$\mathbb{E} \left(\sup_{t \in T} X_t \right) \leq 3 \times \sum_{k \in \mathbb{Z}} 2^{-k} \cdot (\psi^*)^{-1} (2 \log N(T, d, 2^{-k}))$$

(Dudley's integral inequality)

Outline

Introduction

Concentration inequalities

Variance Bounds

- Tensorization property of the Variance
- Bounded Difference Inequalities, Variance
- Poincaré Inequalities

Subgaussian Concentration

- Subgaussianity
- Bounded Difference Inequalities, Subgaussian, weak version
- Tensorization property of the Entropy
- Bounded Difference Inequalities, Subgaussian, strong version
- Log-Sobolev Inequalities

Sub-Exponential Concentration

Suprema

- Finite Maxima
- The Covering Method
- The Chaining Method

Learning Theory

Intro - Learning Theory

Recall classic statistics: parameter estimation (e.g. estimate the mean of a certain distribution)

- ▶ data: $\{x_i\}_{i=1}^n$
- ▶ hypothesis: distribution model of X
- ▶ method: statistics (e.g. $\hat{\theta} = \bar{X}$)
- ▶ analyze/evaluate: properties of the statistics ($\mathbb{E}(\hat{\theta} - \theta)$, ...) with $n/?$

Now in supervised learning: function estimation, i.e. parameter estimation under fixed functional form (e.g.)

- ▶ data: $\{(x_i, y_i)\}_{i=1}^n$
- ▶ hypothesis: distribution model of (X, Y) , $f : \mathcal{X} \rightarrow \mathcal{Y}$, $f \in \mathcal{F}$
- ▶ method: empirical risk minimization or ...

$$\hat{\mathcal{R}}(f) = \frac{1}{n} \sum_{i=1}^n \ell(f(x_i), y_i), \quad \hat{f} = \arg \min_{f \in \mathcal{F}} \hat{\mathcal{R}}(f)$$

- ▶ analyze/evaluate: excess risk with n, m, T

$$\mathcal{R}(\hat{f}) - \mathcal{R}(f^*).$$

More on Learning theory, see xx

Measurements of a function estimation

How well model f approximates pair r.v. $\{(x, y)\}$

Let's first consider the question: how to qualify that **how well a model $f : \mathcal{X} \rightarrow \mathcal{Y}$ approximates the data (distribution of the pair r.v. z)?**

Given (1) the (empirical or population) distribution of a two-dimension pair r.v. $z = (x, y)$, (2) a loss function $\ell : \mathcal{F} \times \mathcal{Z} \rightarrow \mathbb{R}_{\geq 0}$, $\ell(f, z) := \ell(f(x), y)$

► empirical risk

$$\hat{\mathcal{R}}(f) = \frac{1}{n} \sum_{i=1}^n \ell(f, z_i)$$

► population risk

$$\mathcal{R}(f) = \mathbb{E}_{z \sim \mu}[\ell(f, z)]$$

population risk decomposition

$$\mathcal{R}(f) = \hat{\mathcal{R}}(f) + \mathcal{R}(f) - \hat{\mathcal{R}}(f)$$

Examples of loss function

- Mean Squared Loss $\ell(f, (x, y)) = (f(x) - y)^2$
- cross entropy $\ell(f, (x, y)) = -y \log(f(x)) - (1 - y) \log(1 - f(x))$

Note that the population risk equals $\mathbb{E}[\hat{\mathcal{R}}(f)]$ if we consider z to be an $n * 2$ -dimension r.v.[?]

Measurements of a function estimation

How well f_1 approximates f_2

Next: how to qualify **how well f_1 approximates f_2** ?

Moreover given (3) another model $f_2 : \mathcal{X} \rightarrow \mathcal{Y}$

- ▶ difference of population risks

$$\mathcal{R}(f_1) - \mathcal{R}(f_2) = \mathbb{E}_{z \sim \mu}[\ell(f_1, z) - \ell(f_2, z)]$$

Risk Difference Decomposition

$$\begin{aligned}\mathcal{R}(f_1) - \mathcal{R}(f_2) &= [\mathcal{R}(f_1) - \hat{\mathcal{R}}(f_1)] + [\hat{\mathcal{R}}(f_1) - \hat{\mathcal{R}}(f_2)] + [\hat{\mathcal{R}}(f_2) - \mathcal{R}(f_2)] \\ &= \text{gen-gap}(f_1) + [\hat{\mathcal{R}}(f_1) - \hat{\mathcal{R}}(f_2)] - \text{gen-gap}(f_2)\end{aligned}$$

where the generalization gap is defined as

$$\text{gen-gap}(f_1) = \mathcal{R}(f_1) - \hat{\mathcal{R}}(f_1) = \mathbb{E}_z[\ell(f_1, z)] - \frac{1}{n} \sum_{i=1}^n \ell(f_1, z_i).$$

Excess Risk

How well f approximates f^*

Suppose now we only consider function that is in a function class $\mathcal{F}$ (why?) and consider two special function, let f_1 and f_2 be

► $\hat{f}$ by empirical risk minimization $\hat{f} \in \arg \min_{f \in \mathcal{F}} \hat{\mathcal{R}}(f)$ So we have

$$\hat{\mathcal{R}}(\hat{f}) - \hat{\mathcal{R}}(f^*) = \frac{1}{n} \sum_{i=1}^n \ell(\hat{f}(x_i), y_i) - \frac{1}{n} \sum_{i=1}^n \ell(f^*(x_i), y_i) \leq 0$$

► f^* target model

The thing of interest in statistical learning is to analyze the behavior of the difference of population risks, which is called the **Excess Risk**.

$$\begin{aligned} \mathcal{R}(\hat{f}) - \mathcal{R}(f^*) &= \text{gen-gap}(\hat{f}) + [\hat{\mathcal{R}}(\hat{f}) - \hat{\mathcal{R}}(f^*)] - \text{gen-gap}(f^*) \\ &\leq \text{gen-gap}(\hat{f}) - \text{gen-gap}(f^*) \\ &\leq 2 \sup_{f \in \mathcal{F}} |\hat{\mathcal{R}}(f) - \mathcal{R}(f)| \end{aligned}$$

→ uniform bound of gen-gap
(beside uniform bound?)

Uniform bound of gen gap

From previous steps, the question boils down to analyzing the behavior of

$$\sup_{f \in \mathcal{F}} \left| \hat{\mathcal{R}}(f) - \mathcal{R}(f) \right| = \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^n \ell(f, z_i) - \mathbb{E}_{z \sim \mu} [\ell(f, z)] \right|$$

Road Map:

1. Concentration: Consider $\sup_{f \in \mathcal{F}} \left| \hat{\mathcal{R}}(f) - \mathcal{R}(f) \right| = g(Z_1, \dots, Z_n)$, then using concentration inequalities, we can analyze its fluctuations around its mean.
2. Suprema: Define random process $\mathbb{X}_f := \frac{1}{n} \sum_{i=1}^n \ell(f, Z_i) - \mathbb{E}[\ell(f, Z)]$, then using theory of suprema, we can analyze its expected supremum through the properties of the function class.

Expected result:

A bound on $\sup_{f \in \mathcal{F}} \left| \hat{\mathcal{R}}(f) - \mathcal{R}(f) \right|$ in terms of $n, \mathcal{F}$, conditional on μ, ℓ .

Case: binary classification

- ▶ data: $\{(x_i, y_i)\}_{i=1}^n \sim \mu$
- ▶ population: μ over $\mathcal{Z} = \mathcal{X} \times \{0, 1\}$
- ▶ function class: $\mathcal{F} \subseteq \{f : \mathcal{X} \rightarrow \{0, 1\}\}$
- ▶ loss function: for $z = (x, y)$, $\ell(f, z) = \mathbf{1}\{f(x) \neq y\} \in \{0, 1\}$

Since cross entropy loss is bounded, we have

$$\sup_{f \in \mathcal{F}} |\hat{\mathcal{R}}(f) - \mathcal{R}(f)| = g(Z_1, \dots, Z_n), \quad \|D_{ig}\|_{\infty} \leq \frac{2}{n},$$

Then by McDiarmid, for any $t > 0$,

$$\mathbb{P}\{g(Z_1, \dots, Z_n) - \mathbb{E}[g(Z_1, \dots, Z_n)] \geq t\} \leq \exp(-nt^2/2).$$

Equivalently, $\forall \delta \in (0, 1)$,

$$\mathbb{P}(\sup_{f \in \mathcal{F}} |\hat{\mathcal{R}}(f) - \mathcal{R}(f)| \leq \mathbb{E} \left[\sup_{f \in \mathcal{F}} |\hat{\mathcal{R}}(f) - \mathcal{R}(f)| \right] + \sqrt{\frac{2 \log(1/\delta)}{n}}) \geq 1 - \delta$$

Case: binary classification

[Rademacher complexity] The Rademacher complexity of a set $A \subseteq \mathbb{R}^n$

$$\text{Rad}(A) = \mathbb{E}_{\varepsilon} \left(\sup_{\alpha \in A} |\langle \alpha, \varepsilon \rangle| \right)$$

[Symmetrization of Empirical Processes]

$$\begin{aligned} \mathbb{E} \left[\sup_{f \in \mathcal{F}} |\hat{\mathcal{R}}(f) - \mathcal{R}(f)| \right] &\leq \frac{2}{n} \mathbb{E}_{Z, \varepsilon} \left[\sup_{f \in \mathcal{F}} \left| \sum_{i=1}^n \varepsilon_i \ell(f, Z) \right| \right] \\ &= \frac{2}{n} \mathbb{E}_{Z, \varepsilon} \left(\sup_{\alpha \in S(Z)} |\langle \alpha, \varepsilon \rangle| \right) \\ &\leq \frac{2}{n} \mathbb{E}_Z [\text{Rad}(S(Z))] \end{aligned}$$

where

$$S(Z) = \left\{ (\ell(f, Z_1), \dots, \ell(f, Z_n)) : f \in \mathcal{F} \right\} \subseteq [0, 1]^n$$

Case: binary classification

Case: regression

OLS with fixed design

proof of Efron-Stein

Tensorization, martingale method

$$\text{Var}(f) = \mathbb{E}[(f - \mathbb{E}f)^2]$$

$$\mathbb{E}f \rightarrow \underbrace{\mathbb{E}(f|X_1)}_{\Delta_1} \rightarrow \underbrace{\mathbb{E}(f|X_1, X_2)}_{\Delta_2} \rightarrow \dots \mathbb{E}(f|X_1, \dots, X_n) = f$$

$$\text{let } \Delta_k = \mathbb{E}(f|X_1, \dots, X_k) - \mathbb{E}(f|X_1, \dots, X_{k-1})$$

martingale increments

$$\underline{f - \mathbb{E}f} = \sum_{k=1}^n \Delta_k$$

$$\text{so } \text{Var}(f) = \mathbb{E}[(f - \mathbb{E}f)^2] = \mathbb{E}\left[\left(\sum_{k=1}^n \Delta_k\right)^2\right]$$

$$= \sum_{k=1}^n \mathbb{E}[\Delta_k^2] \quad (\mathbb{E}[\Delta_k \Delta_l] = 0) \quad ①$$

And

$$\Delta_k = \mathbb{E}(f|X_1, \dots, X_k) - \mathbb{E}(f|X_1, \dots, X_{k-1})$$

$$= \mathbb{E}[(f - \mathbb{E}(f|X_k)) | X_1, \dots, X_k]$$

$$\Delta_k^2 \stackrel{\text{Jensen}}{\leq} \mathbb{E}[(f - \mathbb{E}(f|X_k))^2 | X_1, \dots, X_k]$$

$$\mathbb{E}[\Delta_k^2] \leq \mathbb{E}[(f - \mathbb{E}(f|X_k))^2]$$

$$= \mathbb{E}[\mathbb{E}[(f - \mathbb{E}(f|X_k))^2 | X_k]]$$

$$= \mathbb{E}[\text{Var}(f|X_k)] \quad ②$$

$$① + ② \Rightarrow \text{Var}(f) \leq \sum_{k=1}^n \mathbb{E}[\text{Var}(f|X_k)]$$

proof of Hoeffding

Assumption: r.v. X takes values in $[a, b]$

$$\text{Let } \psi(\lambda) = \log \mathbb{E} e^{\lambda(X - \mathbb{E}X)}$$

$$\text{NTS: } \psi(\lambda) \leq \frac{\lambda^2(a-b)^2}{8}, \quad \forall \lambda \in \mathbb{R}$$

Taylor expansion

$$\psi(\lambda) = \psi(0) + \psi'(0)(\lambda - 0) + \frac{\psi''(\xi)}{2}(\lambda - 0)^2, \quad \xi \in [0, \lambda]$$

→ bound $\psi(0)$, $\psi'(0)$, $\psi''(\lambda)$

$$\text{w.l.o.g., assume } \mathbb{E}X = 0 \Rightarrow \begin{cases} \psi(\lambda) = \log \mathbb{E} e^{\lambda X} \\ \psi(0) = 0 \end{cases}$$

$$\psi'(\lambda) = \frac{\mathbb{E}(X e^{\lambda X})}{\mathbb{E}(e^{\lambda X})} \quad \psi'(0) = \mathbb{E}X = 0$$

$$\begin{aligned} \psi''(\lambda) &= \frac{\mathbb{E}(X^2 e^{\lambda X}) \mathbb{E}(e^{\lambda X}) - (\mathbb{E}(X e^{\lambda X}))^2}{(\mathbb{E}(e^{\lambda X}))^2} \\ &= \frac{\mathbb{E}(X^2 e^{\lambda X})}{\mathbb{E}(e^{\lambda X})} - \left(\frac{\mathbb{E}(X e^{\lambda X})}{\mathbb{E}(e^{\lambda X})} \right)^2 \end{aligned}$$

exponential
reweighted
variance

$$\text{Define } Q \text{ through } \frac{dQ}{d\lambda} = \frac{e^{\lambda X}}{\mathbb{E}(e^{\lambda X})}$$

$$\psi''(\lambda) = \mathbb{E}_Q X^2 - (\mathbb{E}_Q X)^2 = \text{Var}_Q(X) \quad \text{+trick}$$

$$\leq \frac{(a-b)^2}{4} \quad \text{since } \text{supp}_Q(X) \subseteq [a, b]$$

$$\begin{aligned} \Rightarrow \psi(\lambda) &= 0 + 0 + \frac{\psi''(\xi)}{2}(\lambda - 0)^2 \\ &\leq \frac{\lambda^2(a-b)^2}{8} \end{aligned}$$

$\Rightarrow X$ is $\frac{(a-b)^2}{4}$ -subgaussian

proof of McDiarmid

Let $\mathcal{F}_0 = \{\phi, \omega\}$, $\mathcal{F}_i = \sigma(x_1, \dots, x_i)$

$$\Delta_i = \mathbb{E}(f | \mathcal{F}_i) - \mathbb{E}(f | \mathcal{F}_{i-1})$$

$$\underline{f - \mathbb{E}f} = \sum_{i=1}^n \Delta_i$$

NTS: $\sum_{i=1}^n \Delta_i$ is $\frac{1}{4} \sum_{i=1}^n \|D_i f\|_\infty^2$ - subg

By Azuma-Hoeffding, we can construct $a_i = D_i \mathbb{E}f$
that $|b_i - a_i|$ is bounded by $\|D_i f\|_\infty \forall i$

$$\Delta_i \leq \mathbb{E}(\sup_{x_i} f | \mathcal{F}_i) - \mathbb{E}(f | \mathcal{F}_{i-1})$$

$$= \mathbb{E}(\sup_{x_i} f | \mathcal{F}_{i-1}) - \mathbb{E}(f | \mathcal{F}_{i-1})$$

$$= \mathbb{E}[(\sup_{x_i} f - f) | \mathcal{F}_{i-1}] \triangleq b_i$$

$$\Delta_i \geq \mathbb{E}[(f - \inf_{x_i} f) | \mathcal{F}_{i-1}] \triangleq a_i$$

$$\begin{aligned} \text{so } |b_i - a_i| &= |\mathbb{E}[\sup_{x_i} f - \inf_{x_i} f | \mathcal{F}_{i-1}]| \\ &= |\mathbb{E}[D_i f | \mathcal{F}_{i-1}]| \leq \|D_i f\|_\infty \end{aligned}$$

References

- Handel, Ramon van (2016). “Probability in High Dimension”. URL: <https://web.math.princeton.edu/~rvan/APC550.pdf>.
- Vershynin, Roman (2026). *High-Dimensional Probability: An Introduction with Applications in Data Science*. 2nd ed. Cambridge Series in Statistical and Probabilistic Mathematics. Cambridge: Cambridge University Press.
- Wang, Kaizheng (2026). “Pseudo-labeling for kernel ridge regression under covariate shift”. In: *The Annals of Statistics* 54.1, pp. 252–276.
- Wu, Lei (2026). *Topics in Deep Learning Theory*. Lecture Notes, Peking University. Accessed: 2026-06-29. URL: <https://leiwu0.github.io/courses/dl-theory/index.html>.